

# Mutual potential between two rigid bodies with arbitrary shapes and mass distributions

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**Abstract** Formulae to compute the mutual potential, force, and torque between two rigid bodies are given. These formulae are expressed in Cartesian coordinates using inertia integrals. They are valid for rigid bodies with arbitrary shapes and mass distributions. By using recursive relations, these formulae can be easily implemented on computers. Comparisons with previous studies show their superiority in computation speed. Using the algorithm as a tool, the planar problem of two ellipsoids is studied. Generally, potential truncated at the second order is good enough for a qualitative description of the mutual dynamics. However, for ellipsoids with very large non-spherical terms, higher order terms of the potential should be considered, at the cost of a higher computational cost. Explicit formulae of the potential truncated to the fourth order are given.

**Keywords** Full two rigid body problem · Binary systems · Spin–orbit coupling · Numerical simulations · Stability

## 1 Introduction

The full two rigid body problem (F2RBP) is a problem of current and historical interest in Celestial Mechanics. Compared with the simple and integrable two body problem, it is difficult and generally non-integrable. This problem can be used to describe many practical systems such as binary asteroids (Margot et al. 2015; Pravec et al. 2010; Chappaz and Howell 2015; Ferrari et al. 2016), binary stars (Carvalho et al. 2016), and attitude dynamics of satellites (Wang and Xu 2016). Due to limitations of analytical approaches to the problem (Wisdom and Peale 1984; Giacaglia and Jefferys 1971; Kinoshita 1972; Scheeres 2006),

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numerical approaches for its study are often taken, especially when the two bodies are very close to each other.

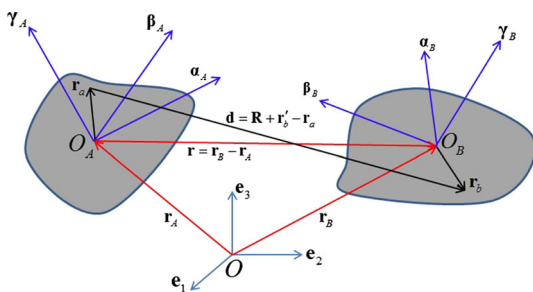
A basic requirement for the study of the F2RBP is knowledge of the mutual potential between the two rigid bodies. The usual convention to express the non-spherical gravitational potential of a body is by use of spherical harmonics. In the case of the F2RBP, such expressions are not as simple as for a single non-spherical body, due to the necessity of tracking the relative attitude between the two rigid bodies and the need to account for the coupling between higher order expansion terms of each body's harmonic expansion. [Giacaglia and Jefferys \(1971\)](#) gave an explicit expansion to the 4th order. Using a rotational transformation theorem given by [Wigner \(1959\)](#), [Borderies \(1978\)](#) was able to carry on the expansion to infinite orders. By considering the relative attitude between the two rigid bodies and expressing the potential in the body-fixed frame of one body, [Maciejewski \(1995\)](#) simplified the expression given by Borderies. An equivalent way to the spherical harmonic approach is given by [Hartmann et al. \(1994\)](#) using the symmetric trace-free (STF) tensor. Using this same approach, a recent paper by [Compère and Lemaître \(2014\)](#) studied the binary asteroid system 1999KW4 (66391) which was previously studied by other authors via the mutual polyhedron potential approach ([Scheeres et al. 2006](#); [Fahnestock and Scheeres 2008](#)).

For numerical integrations in Cartesian coordinates, spherical harmonics are not always an optimal choice. In this paper we take a more direct approach to expand the potential in Cartesian coordinates, replacing the Stokes coefficients with the so-called high order inertia integrals. [Schutz \(1979\)](#) gave explicit expressions to the 4th order (by neglecting the 3rd order terms) for the potential and to the 2nd order for the torque, although these expressions were incomplete. [Ashenberg \(2007\)](#) completed the expressions and extended the work to the 4th order for both the potential and the torque. [Naidu and Margot \(2015\)](#) used Ashenberg's expressions to study the libration and chaotic rotation of the secondary in a binary system composed of two ellipsoids. A very similar approach was taken by [Paul \(1988\)](#) to expand the potential to infinite orders. [Tricarico \(2008\)](#) completed that work by giving explicit formulae for the force and torque. A drawback of both Paul's (and Tricarico's) formulae are that there are too many open summations.

An alternate approach to the F2RBP is the polyhedron mutual potential ([Werner and Scheeres 2005](#); [Fahnestock and Scheeres 2006](#); [Hirabayashi and Scheeres 2013](#)). This approach makes intuitive use of the polyhedron model for asteroids. One issue with this approach is that the shape information and the relative attitude of the two bodies are combined together. At every step of the numerical integration, the potential (also the force and the torque) should be completely recomputed over the two bodies. Computational time is another issue as the computation must be done over all the tetrahedron pairs between the two bodies. This significantly increases the computation burden when polyhedrons with many faces are used.

In this contribution, we propose a new numerical approach to computing the mutual potential in the F2RBP. The results are valid for rigid bodies with arbitrary shapes (polyhedrons, ellipsoids, etc.) and mass distributions. Our results are qualitatively equivalent to those given by [Paul \(1988\)](#), but there is only one open summation in our expressions. Due to the use of recurrence relations, our approach can be easily realized on computers (see the discussion part of the paper). This new approach is compared with previous methods. As far as the authors know, although many algorithms about the mutual potential in the F2RBP have been proposed, numerical simulations available in literature are only done for the polyhedron approach and the STF tensor approach. As a result, we can only compare our results with those two. The comparisons show that our approach is much faster than both, and the advantage becomes more obvious as higher order expansions are used.

**Fig. 1** Geometry of the two rigid bodies with respect to each other and the inertial frame



Using our computation algorithm as a tool, the planar F2RBP composed of two ellipsoids is studied, with special emphasis on the fission case. This model was previously used to simulate the fission process of contact binaries (Scheeres 2007, 2009) and to determine the evolution of binary asteroids under tidal dissipations (Jacobson and Scheeres 2011). In this contribution, results corresponding to potentials truncated at different orders are compared. Our studies show that potential truncated at the 2nd order is generally good enough for a qualitative description of the mutual dynamics. However, for ellipsoids with a very large  $J_{22}$  term, fourth or even higher order terms of the potential should be considered. Explicit formulae of the 4th order potential are given in the appendix.

## 2 Equations of motion

The geometry of the F2RBP in the inertial frame is depicted in Fig. 1.  $O_A$  and  $O_B$  are respectively the barycenters of the two rigid bodies.  $(\alpha_A, \beta_A, \gamma_A; O_A)$  and  $(\alpha_B, \beta_B, \gamma_B; O_B)$  are their respective body-fixed frames.  $(e_1, e_2, e_3; O)$  is the inertial frame.

The equations of Motion (EOM) given by Maciejewski (1995) are used in the work. Expressed in the body-fixed frame of A, they have the following form

$$\begin{cases} \dot{\mathbf{R}} = \mathbf{R} \times \boldsymbol{\Omega}_A + \mathbf{V}, & \dot{\mathbf{V}} = \mathbf{V} \times \boldsymbol{\Omega}_A - \frac{1}{m} \frac{\partial U}{\partial \mathbf{R}} \\ \dot{\mathbf{L}}_A = \mathbf{L}_A \times \boldsymbol{\Omega}_A + \mathbf{M}_A, & \dot{\mathbf{L}}'_B = \mathbf{L}'_B \times \boldsymbol{\Omega}_A + \mathbf{M}'_B \end{cases} \quad (1)$$

where  $U$  is the potential energy between the two rigid bodies, and

$$\mathbf{R} = C_A^T \mathbf{r}, \quad \mathbf{V} = C_A^T \dot{\mathbf{r}}, \quad m = \frac{m_A m_B}{m_A + m_B}$$

The matrix  $C_A$  (similarly  $C_B$ ) transforms a vector from the body fixed frame of A(B) to the inertial frame. They are

$$C_A = [\alpha_A, \beta_A, \gamma_A], \quad C_B = [\alpha_B, \beta_B, \gamma_B]$$

$\mathbf{L}'_B$  and  $\mathbf{M}'_B$  are respectively the rotational angular momentum of the body B and the torque it receives from the body A. The prime means that they are expressed in the body fixed frame of A. The two angular speeds are computed by

$$\boldsymbol{\Omega}_A = \mathbf{I}_A^{-1} \mathbf{L}_A, \quad \boldsymbol{\Omega}_B = \mathbf{I}_B^{-1} C^T \mathbf{L}'_B$$

The torques  $\mathbf{M}_A, \mathbf{M}'_B$  can be computed by the following formulae

$$\mathbf{M}'_B = -\alpha \times \frac{\partial U}{\partial \alpha} - \beta \times \frac{\partial U}{\partial \beta} - \gamma \times \frac{\partial U}{\partial \gamma}, \quad \mathbf{M}_A = \mathbf{R} \times \frac{\partial U}{\partial \mathbf{R}} - \mathbf{M}'_B \quad (2)$$

where the transformation matrix

$$C = [\alpha, \beta, \gamma] = C_A^T C_B$$

transforms a vector from the body-fixed frame of  $B$  to the body-fixed frame of  $A$ . Usually  $C_B$  is not necessary unless we need to individually know the attitude of the body  $B$  in the inertial frame, thus only the matrixes  $C_A^T$  and  $C$  are computed. Their EOMs satisfy

$$\dot{C} = C\tilde{\Omega}_B - \tilde{\Omega}_A C, \quad \dot{C}_A = C_A \tilde{\Omega}_A \quad (3)$$

where

$$\tilde{\Omega}_* = \begin{bmatrix} 0 & -\Omega_*^z & \Omega_*^y \\ \Omega_*^z & 0 & -\Omega_*^x \\ -\Omega_*^y & \Omega_*^x & 0 \end{bmatrix}.$$

For our specific numerical computations we use Euler parameters (quaternions) instead of Eq. (3), the details are standard and omitted here.

### 3 Potential, force, and torque

Obviously, to integrate Eq. (1), the following four quantities are necessary

$$\frac{\partial U}{\partial \mathbf{R}}, \quad \frac{\partial U}{\partial \alpha}, \quad \frac{\partial U}{\partial \beta}, \quad \frac{\partial U}{\partial \gamma}$$

In the following, we'll describe in detail how to get these terms. The potential energy is

$$U = -G \int_A \int_B \frac{1}{d} dm_A dm_B \quad (4)$$

From Fig. 1 we have

$$d^2 = R^2 + h^2 - 2\mathbf{R} \cdot \mathbf{h}, \quad \mathbf{h} = \mathbf{r}_a - \mathbf{r}'_b \quad (5)$$

where the prime means the vector is expressed in the body-fixed frame of  $A$ . Define

$$\chi_n = \frac{1}{R} \left( \frac{h}{R} \right)^n P_n \left( \frac{\mathbf{R} \cdot \mathbf{h}}{Rh} \right) \quad (6)$$

We have

$$\frac{1}{d} = \frac{1}{R} \sum_{n=0}^{\infty} \left( \frac{h}{R} \right)^n P_n \left( \frac{\mathbf{R} \cdot \mathbf{h}}{Rh} \right) = \sum_{n=0}^{\infty} \chi_n \quad (7)$$

It can be shown that  $\chi_n$  can be expressed as

$$\chi_n = \frac{1}{R^{n+1}} \sum_{k(2)=n-2[\frac{n}{2}]}^n t_k^n \left( \frac{\mathbf{R}}{R} \cdot \mathbf{h} \right)^k (h^2)^{\frac{n-k}{2}} \quad (8)$$

where  $k(2)$  means the step size in the summation is 2 instead of 1, and  $[\frac{n}{2}]$  means the integer part of  $\frac{n}{2}$ . Then the following recursive formula exists (Wang and Guo 1989)

$$t_{k+2}^n = -\frac{(n-k)(n+k+1)}{(k+2)(k+1)} t_k^n \quad (9)$$

where

$$t_0^n = (-1)^{\frac{n}{2}} \frac{n!}{2^n (\frac{n}{2}!)^2} \text{ (when } n \text{ is even)}, \quad t_1^n = (-1)^{\frac{n-1}{2}} \frac{n!}{2^{n-1} (\frac{n-1}{2}!)^2} \text{ (when } n \text{ is odd)}$$

Define

$$\mathbf{e} = \frac{\mathbf{R}}{R} = (e_x, e_y, e_z)^T$$

We can further expand the two factors in Eq. (8) as

$$\left(\frac{\mathbf{R}}{R} \cdot \mathbf{h}\right)^k = \sum_{(i_1 i_2 i_3)(i_4 i_5 i_6)} a_{(i_1 i_2 i_3)(i_4 i_5 i_6)}^k (e_x x_a)^{i_1} (e_y y_a)^{i_2} (e_z z_a)^{i_3} (e_x x'_b)^{i_4} (e_y y'_b)^{i_5} (e_z z'_b)^{i_6} \quad (10)$$

$$(h^2)^{\frac{n-k}{2}} = \sum_{(j_1 j_2 j_3)(j_4 j_5 j_6)} b_{(j_1 j_2 j_3)(j_4 j_5 j_6)}^{n-k} (x_a)^{j_1} (y_a)^{j_2} (z_a)^{j_3} (x'_b)^{j_4} (y'_b)^{j_5} (z'_b)^{j_6} \quad (11)$$

where the following restrictions hold

$$k = i_1 + i_2 + i_3 + i_4 + i_5 + i_6, \quad n - k = j_1 + j_2 + j_3 + j_4 + j_5 + j_6 \quad (12)$$

and  $a_{(i_1 i_2 i_3)(i_4 i_5 i_6)}^k = 0$  and  $b_{(j_1 j_2 j_3)(j_4 j_5 j_6)}^{n-k} = 0$  if one of the subscripts is smaller than 0. The recursive formulae for the coefficients are

$$\begin{aligned} a_{(i_1 i_2 i_3)(i_4 i_5 i_6)}^k &= a_{(i_1-1 i_2 i_3)(i_4 i_5 i_6)}^{k-1} + a_{(i_1 i_2-1 i_3)(i_4 i_5 i_6)}^{k-1} + a_{(i_1 i_2 i_3-1)(i_4 i_5 i_6)}^{k-1} \\ &\quad - a_{(i_1 i_2 i_3)(i_4-1 i_5 i_6)}^{k-1} - a_{(i_1 i_2 i_3)(i_4 i_5-1 i_6)}^{k-1} - a_{(i_1 i_2 i_3)(i_4 i_5 i_6-1)}^{k-1} \\ b_{(j_1 j_2 j_3)(j_4 j_5 j_6)}^k &= b_{(j_1-2 j_2 j_3)(j_4 j_5 j_6)}^{k-2} + b_{(j_1 j_2-2 j_3)(j_4 j_5 j_6)}^{k-2} + b_{(j_1 j_2 j_3-2)(j_4 j_5 j_6)}^{k-2} \\ &\quad + b_{(j_1 j_2 j_3)(j_4-2 j_5 j_6)}^{k-2} + b_{(j_1 j_2 j_3)(j_4 j_5-2 j_6)}^{k-2} + b_{(j_1 j_2 j_3)(j_4 j_5 j_6-2)}^{k-2} \\ &\quad - 2b_{(j_1-1 j_2 j_3)(j_4-1 j_5 j_6)}^{k-2} - 2b_{(j_1 j_2-1 j_3)(j_4 j_5-1 j_6)}^{k-2} - 2b_{(j_1 j_2 j_3-1)(j_4 j_5 j_6-1)}^{k-2} \end{aligned} \quad (13)$$

The initial values to start the recursive process are

$$\begin{aligned} a_{(000)(000)}^0 &= 1, \\ a_{(100)(000)}^1 &= 1, \quad a_{(010)(000)}^1 = 1, \quad a_{(001)(000)}^1 = 1, \\ a_{(000)(100)}^1 &= -1, \quad a_{(000)(010)}^1 = -1, \quad a_{(000)(001)}^1 = -1 \\ b_{(000)(000)}^0 &= 1, \\ b_{(200)(000)}^2 &= 1, \quad b_{(020)(000)}^2 = 1, \quad b_{(002)(000)}^2 = 1 \\ b_{(000)(200)}^2 &= 1, \quad b_{(000)(020)}^2 = 1, \quad b_{(000)(002)}^2 = 1 \\ b_{(100)(100)}^2 &= -2, \quad b_{(010)(010)}^2 = -2, \quad b_{(001)(001)}^2 = -2 \end{aligned}$$

These coefficients are constants and can be pre-computed and stored in the program. Substituting Eqs. (10) and (11) into Eq. (8), we have

$$\begin{aligned} \chi_n = & \frac{1}{R^{n+1}} \sum_{k(2)=n-2[\frac{n}{2}]}^n t_k^n \sum_{(i_1 i_2 i_3)(i_4 i_5 i_6)(j_1 j_2 j_3)(j_4 j_5 j_6)} a_{(i_1 i_2 i_3)(i_4 i_5 i_6)}^k b_{(j_1 j_2 j_3)(j_4 j_5 j_6)}^{n-k} \\ & \times e_x^{i_1+i_4} e_y^{i_2+i_5} e_z^{i_3+i_6} (x_a)^{i_1+j_1} (y_a)^{i_2+j_2} (z_a)^{i_3+j_3} (x'_b)^{i_4+j_4} (y'_b)^{i_5+j_5} (z'_b)^{i_6+j_6} \end{aligned} \quad (14)$$

Considering Eqs. (4) and (7), the potential energy is

$$U = -G \sum_{n=0}^{\infty} \int_A \int_B \chi_n dm_A dm_B \quad (15)$$

Substituting Eq. (14) in and defining the generalized inertia integrals as

$$T_A^{lmn} = \int_A (x_a)^l (y_a)^m (z_a)^n dm_A, \quad T_B'^{lmn} = \int_B (x'_b)^l (y'_b)^m (z'_b)^n dm_B \quad (16)$$

We get the final formulae for the potential energy

$$U = -G \sum_{n=0}^{\infty} \frac{1}{R^{n+1}} \tilde{U}_n \quad (17)$$

where

$$\begin{aligned} \tilde{U}_n = & \sum_{k(2)=n-2[\frac{n}{2}]}^n t_k^n \sum_{(i_1 i_2 i_3)(i_4 i_5 i_6)(j_1 j_2 j_3)(j_4 j_5 j_6)} a_{(i_1 i_2 i_3)(i_4 i_5 i_6)}^k b_{(j_1 j_2 j_3)(j_4 j_5 j_6)}^{n-k} \\ & e_x^{i_1+i_4} e_y^{i_2+i_5} e_z^{i_3+i_6} T_A^{(i_1+j_1)(i_2+j_2)(i_3+j_3)} T_B'^{(i_4+j_4)(i_5+j_5)(i_6+j_6)} \end{aligned} \quad (18)$$

For the two terms defined in Eq. (16),  $T_A^{lmn}$  is simple to compute as all items in the integral are constants in the body-fixed frame of  $A$ . But for the term  $T_B'^{lmn}$ , the coordinates change with the relative altitude of  $B$  with respect to  $A$ . Denote the transformation matrix from the body fixed frame of  $B$  to that of  $A$  as

$$C = [\alpha, \beta, \gamma] = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix}. \quad (19)$$

we have

$$x'_b = c_{11}x_b + c_{12}y_b + c_{13}z_b, \quad y'_b = c_{21}x_b + c_{22}y_b + c_{23}z_b, \quad z'_b = c_{31}x_b + c_{32}y_b + c_{33}z_b$$

As a result, we have

$$\begin{aligned} (x'_b)^l (y'_b)^m (z'_b)^n = & \sum_{i_1=0}^l \sum_{j_1=0}^{l-i_1} \sum_{i_2=0}^m \sum_{j_2=0}^{m-i_2} \sum_{i_3=0}^n \sum_{j_3=0}^{n-i_3} \\ & \times \frac{l!}{i_1!j_1!(l-i_1-j_1)!} \frac{m!}{i_2!j_2!(m-i_2-j_2)!} \frac{n!}{i_3!j_3!(n-i_3-j_3)!} \\ & \times c_{11}^{i_1} c_{12}^{j_1} c_{13}^{l-i_1-j_1} c_{21}^{i_2} c_{22}^{j_2} c_{23}^{m-i_2-j_2} c_{31}^{i_3} c_{32}^{j_3} c_{33}^{n-i_3-j_3} (x_b)^{i_1+i_2+i_3} (y_b)^{j_1+j_2+j_3} \\ & \times (z_b)^{l+m+n-i_1-i_2-i_3-j_1-j_2-j_3} \end{aligned}$$

Define

$$T_B^{lmn} = \int_B (x_b)^l (y_b)^m (z_b)^n dm_B \quad (20)$$

We have

$$T_B^{lmn} = \sum_{i_1=0}^l \sum_{j_1=0}^{l-i_1} \sum_{i_2=0}^m \sum_{j_2=0}^{m-i_2} \sum_{i_3=0}^n \sum_{j_3=0}^{n-i_3} \frac{l!}{i_1!j_1!(l-i_1-j_1)!} \frac{m!}{i_2!j_2!(m-i_2-j_2)!} \frac{n!}{i_3!j_3!(n-i_3-j_3)!} \\ \times c_{11}^{i_1} c_{12}^{j_1} c_{13}^{i_1+j_1} c_{21}^{i_2} c_{22}^{j_2} c_{23}^{i_2+j_2} c_{31}^{i_3} c_{32}^{j_3} c_{33}^{i_3+j_3} T_B^{(i_1+i_2+i_3)(j_1+j_2+j_3) \times (l+m+n-i_1-i_2-i_3-j_1-j_2-j_3)} \quad (21)$$

where  $T_B^{lmn}$  defined in Eq. (20) is in the body-fixed frame of  $B$ . For fixed bodies  $A$  and  $B$ , the terms  $T_B^{lmn}$  and  $T_B^{lmn}$  are constants. At every step of the integration,  $T_B^{lmn}$  can be computed from  $T_B^{lmn}$  by using Eq. (21).

The derivative of the potential with respect to  $\mathbf{R}$  can be directly inferred from Eqs. (17) and (18). Notice that  $\frac{\partial e_x}{\partial \mathbf{R}}$ ,  $\frac{\partial e_y}{\partial \mathbf{R}}$ ,  $\frac{\partial e_z}{\partial \mathbf{R}}$  should also be considered when taking the derivatives. We have the following formulae.

$$-\frac{\partial U}{\partial (*)} = G \sum_{n=0}^{\infty} \left( -\frac{(n+1)(*)}{R^{n+3}} \tilde{U}_n + \frac{1}{R^{n+1}} \frac{\partial \tilde{U}_n}{\partial (*)} \right) \quad (22)$$

where  $(*)$  represents  $X$ ,  $Y$  or  $Z$ , and

$$\frac{\partial \tilde{U}_n}{\partial (*)} = \sum_{k(2)=n-2[\frac{n}{2}]}^n t_k^n \sum_{(i_1 i_2 i_3)(i_4 i_5 i_6)(j_1 j_2 j_3)(j_4 j_5 j_6)} a_{(i_1 i_2 i_3)(i_4 i_5 i_6)}^k b_{(j_1 j_2 j_3)(j_4 j_5 j_6)}^{n-k} \\ T_A^{(i_1+j_1)(i_2+j_2)(i_3+j_3)} T_B^{(i_4+j_4)(i_5+j_5)(i_6+j_6)} \left[ (i_1+i_4) e_x^{i_1+i_4-1} e_y^{i_2+i_5} e_z^{i_3+i_6} \frac{\partial e_x}{\partial (*)} \right. \\ \left. + (i_2+i_5) e_x^{i_1+i_4} e_y^{i_2+i_5-1} e_z^{i_3+i_6} \frac{\partial e_y}{\partial (*)} + (i_3+i_6) e_x^{i_1+i_4} e_y^{i_2+i_5} e_z^{i_3+i_6-1} \frac{\partial e_z}{\partial (*)} \right]$$

One remark is that if the index of the terms in the above formulae is less than 0, they have no contribution and should be removed from the summation.

The derivatives of the potential with respect to  $\alpha$ ,  $\beta$ ,  $\gamma$  which is necessary in computing the torque can also be easily computed from Eqs. (18) and (21). We have the following formulae

$$-\frac{\partial U}{\partial c_{ij}} = G \sum_{n=0}^{\infty} \frac{1}{R^{n+1}} \frac{\partial \tilde{U}_n}{\partial c_{ij}} \\ \frac{\partial \tilde{U}_n}{\partial c_{ij}} = \sum_{k(2)=n-2[\frac{n}{2}]}^n t_k^n \sum_{(i_1 i_2 i_3)(i_4 i_5 i_6)(j_1 j_2 j_3)(j_4 j_5 j_6)} a_{(i_1 i_2 i_3)(i_4 i_5 i_6)}^k b_{(j_1 j_2 j_3)(j_4 j_5 j_6)}^{n-k} \\ e_x^{i_1+i_4} e_y^{i_2+i_5} e_z^{i_3+i_6} T_A^{(i_1+j_1)(i_2+j_2)(i_3+j_3)} \frac{\partial T_B^{(i_4+j_4)(i_5+j_5)(i_6+j_6)}}{\partial c_{ij}} \quad (23)$$

where the last term  $\frac{\partial T_B^{lmn}}{\partial c_{ij}}$  can be easily obtained from Eq. (21).

## 4 Generalized inertia integrals

To implement the formulae in the above section, we need to compute the integrals  $T_A^{lmn}$  and  $T_B^{lmn}$ . For rigid bodies, they are constants. If we have accurate information on the body's shape and its mass distribution, they can be computed. They can also be computed from the Stokes coefficients or vice versa (Tricarico 2008). For the purpose of comparing the polyhedron approach with the STF tensor approach, we describe how to compute these quantities for the polyhedron model. Suppose the asteroid is decomposed as  $n$  tetrahedrons, with one common vertex at the barycenter. For each tetrahedron  $a$ , suppose the coordinates of the other three vertices are

$$(x_1^a, y_1^a, z_1^a), \quad (x_2^a, y_2^a, z_2^a), \quad (x_3^a, y_3^a, z_3^a)$$

The following transformation goes between the body fixed frame of  $A$  and the "standard" tetrahedron which has one vertex at the origin and the other three vertices respectively at  $(1,0,0)$ ,  $(0,1,0)$ , and  $(0,0,1)$ .

$$\mathbf{r}_a = \begin{pmatrix} x_1^a & x_2^a & x_3^a \\ y_1^a & y_2^a & y_3^a \\ z_1^a & z_2^a & z_3^a \end{pmatrix} \begin{pmatrix} u_a \\ v_a \\ w_a \end{pmatrix} \quad (24)$$

Similar to Eq. (21), we get

$$\begin{aligned} x_a^l y_a^m z_a^n &= \sum_{i_1=0}^l \sum_{j_1=0}^{l-i_1} \sum_{i_2=0}^m \sum_{j_2=0}^{m-i_2} \sum_{i_3=0}^n \sum_{j_3=0}^{n-i_3} \\ &\quad \times \frac{l!}{i_1! j_1! (l-i_1-j_1)!} \frac{m!}{i_2! j_2! (m-i_2-j_2)!} \frac{n!}{i_3! j_3! (n-i_3-j_3)!} \\ &\quad \times (x_1^a)^{i_1} (x_2^a)^{j_1} (x_3^a)^{l-i_1-j_1} (y_1^a)^{i_2} (y_2^a)^{j_2} (y_3^a)^{m-i_2-j_2} (z_1^a)^{i_3} (z_2^a)^{j_3} (z_3^a)^{n-i_3-j_3} \\ &\quad \times u_a^{i_1+i_2+i_3} v_a^{j_1+j_2+j_3} w_a^{l+m+n-i_1-i_2-i_3-j_1-j_2-j_3} \end{aligned}$$

Denote

$$Q_a^{ijk} = \int_a u_a^i v_a^j w_a^k du_a dv_a dw_a = \frac{i! j! k!}{(i+j+k+3)!}$$

We have

$$\begin{aligned} T_A^{lmn} &= \sum_a \rho_a T_a \sum_{i_1=0}^l \sum_{j_1=0}^{l-i_1} \sum_{i_2=0}^m \sum_{j_2=0}^{m-i_2} \sum_{i_3=0}^n \sum_{j_3=0}^{n-i_3} \\ &\quad \times \frac{l!}{i_1! j_1! (l-i_1-j_1)!} \frac{m!}{i_2! j_2! (m-i_2-j_2)!} \frac{n!}{i_3! j_3! (n-i_3-j_3)!} \\ &\quad \times (x_1^a)^{i_1} (x_2^a)^{j_1} (x_3^a)^{l-i_1-j_1} (y_1^a)^{i_2} (y_2^a)^{j_2} (y_3^a)^{m-i_2-j_2} (z_1^a)^{i_3} (z_2^a)^{j_3} (z_3^a)^{n-i_3-j_3} \\ &\quad \times Q_a^{(i_1+i_2+i_3)(j_1+j_2+j_3)(l+m+n-i_1-i_2-i_3-j_1-j_2-j_3)} \end{aligned} \quad (25)$$

where  $\rho_a$  is the density of the tetrahedron and  $T_a$  is the Jacobian matrix generated by the coordinate transformation Eq. (24). The quantity  $T_B^{lmn}$  can be computed in the same way.

The algorithm described above is relatively direct to implement, and although it seems that there are too many summations in Eq. (18), actually it is just an explicit expression of the Legendre expansion Eq. (7) and when the series is truncated at a low order, there are only a few terms. The shape information (for inhomogeneous bodies, also including the density



information) of the body  $B$  is recorded in  $T_B^{lmn}$ . The attitude information of the body  $B$  with respect to  $A$  is recorded in the matrix  $C$ . At every integration step, the shape information and the attitude information are multiplied by Eq. (21) to get  $T_B^{lmn}$ . Since the quantities  $T_A^{lmn}$ ,  $T_B^{lmn}$  and the coefficients  $a_k, b_k, t_k^n$  can be computed beforehand, the main computation task at every step is Eq. (21).

## 5 Numerical simulations

To demonstrate the efficiency of the above algorithm, two numerical simulations are done. The first one is for a simple binary system composed of two octahedrons. By comparison with the polyhedron model, the accuracy of the program is demonstrated and its superiority is analyzed. The second simulation is for the binary asteroid system 1999 KW4 (66391). The computation efficiency is compared with both the polyhedron approach (Scheeres et al. 2006; Fahnestock and Scheeres 2008) and the STF tensor approach (Compère and Lemaître 2014).

### 5.1 Two octahedrons

Two octahedrons with homogenous densities  $2 \times 10^3 \text{ kg m}^{-3}$  are used. The position vectors of the six vertices in their body-fixed frames are

$$\begin{cases} P_{+x}^A = (1, 0, 0)m - \mathbf{B}_A, & P_{-x}^A = (-2, 0, 0)m - \mathbf{B}_A \\ P_{+y}^A = (0, 1/\sqrt{2}, 0)m - \mathbf{B}_A, & P_{-y}^A = (0, -\sqrt{2}, 0)m - \mathbf{B}_A \\ P_{+z}^A = (0, 0, 1)m - \mathbf{B}_A, & P_{-z}^A = (0, 0, -\sqrt{2})m - \mathbf{B}_A \end{cases}$$

$$\begin{cases} P_{+x}^B = (\sqrt{2}, 0, 0)m - \mathbf{B}_B, & P_{-x}^B = (-1, 0, 0)m - \mathbf{B}_B \\ P_{+y}^B = (0, \sqrt{3}, 0)m - \mathbf{B}_B, & P_{-y}^B = (0, -\sqrt{3}/2, 0)m - \mathbf{B}_B \\ P_{+z}^B = (0, 0, \sqrt{5})m - \mathbf{B}_B, & P_{-z}^B = (0, 0, -1)m - \mathbf{B}_B \end{cases}$$

where the two vectors

$$\begin{aligned} \mathbf{B}_A &= (-0.250000000000000, -0.176776695296637, -0.103553390593274)m, \\ \mathbf{B}_B &= (0.103553390593274, 0.216506350946110, 0.309016994374947)m \end{aligned}$$

are introduced to make sure that the origin of the body-fixed frame is the barycenter of the body A (or B). The position vector  $\mathbf{r}$  and the speed vector  $\dot{\mathbf{r}}$  are given by the two-body relation using the following Keplerian orbital elements.

$$a = 5 \text{ m}, \quad e = 0.0, \quad i = 30^\circ, \quad \Omega = 30^\circ, \quad \omega = 30^\circ, \quad M = 0^\circ$$

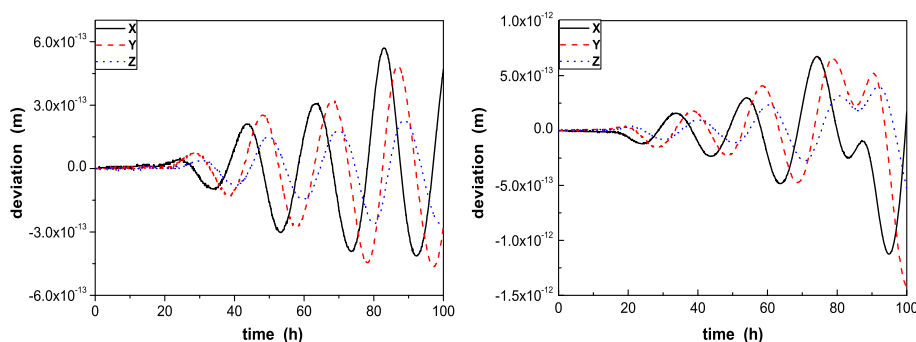
The initial three (3–1–3) Euler angles of the body-fixed frame of A (or B) with respect to the inertial frame are

$$\psi_A = 30^\circ, \theta_A = 10^\circ, \phi_A = 60^\circ, \quad \psi_B = 15^\circ, \theta_B = 0^\circ, \phi_B = 45^\circ$$

The initial angular speeds of the two bodies in their body-fixed frames are

$$\Omega_A = (0.1, 0.3, 0.8)^T \omega_0, \quad \Omega_B = (0.1, 0.2, 1.2)^T \omega_0$$

where  $\omega_0$  is the orbital angular velocity of the Keplerian orbit given by the above parameters. A program using the above algorithm has been compiled in FORTRAN. The verification of the program is demonstrated by two criteria: conservation of the total system energy and



**Fig. 2** Deviations between the orbit integrated by the above algorithm and the polyhedron approach, for an integration time of 100h. *Left* 3rd solution; *right* 5th order solution

**Table 1** Computation time for the polyhedron approach and our approach

| Order $n$ | $t_p$ (s) | $t_e$ (s) | $t_p/t_e$ |
|-----------|-----------|-----------|-----------|
| 2         | 1.2792    | 0.0056    | 228.4286  |
| 3         | 10.4989   | 0.0194    | 541.1804  |
| 4         | 75.7697   | 0.0630    | 1202.6937 |
| 5         | 529.2958  | 0.1920    | 2756.7490 |

momentum (details are omitted), and comparison with the polyhedron approach. Since the above algorithm and the polyhedron approach (Werner and Scheeres 2005) are both based on the Legendre expansion Eq. (7), they should give same results for potentials truncated at the same order  $n$ . Taking  $n = 3$  and  $n = 5$  as examples, Fig. 2 shows the time history of the deviations between the orbit integrated by the above algorithm and the polyhedron approach. For both approaches, a RKF7(8) integrator is used, with a fixed step size of 300 s. Considering the accuracy of numerical integrations, the two orbits can be taken as identical.

After demonstrating the correctness of the program, we compare the computational efficiencies of the algorithms. Table 1 shows the CPU time needed for the polyhedron approach (2nd column) and our approach (3rd column) for 100 steps. The last column is the ratio between the two. The 1st column is the truncation order of the potential. A single processor with a CPU frequency of 2.9 GHz is used. The superiority of our approach over the polyhedron approach is obvious.

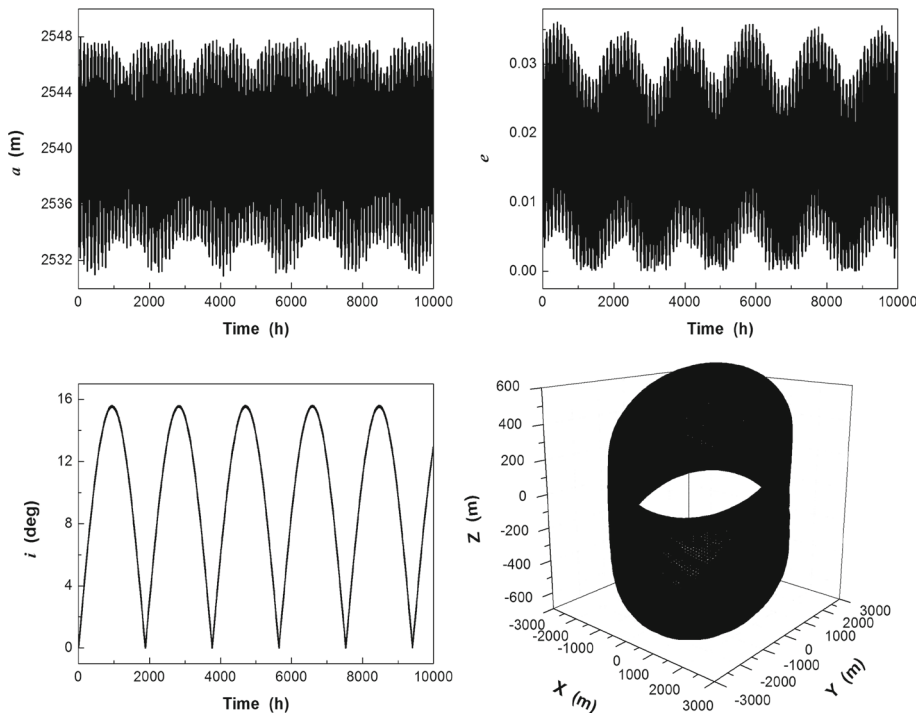
## 5.2 1999 KW4

We also test the algorithm on the binary asteroid pair 1999 KW4 (66391). It has been studied respectively via the polyhedron approach (Fahnestock and Scheeres 2008) and the STF tensor approach (Compère and Lemaître 2014). The shape models of the two binaries can be downloaded from the website (<http://echo.jpl.nasa.gov/asteroids/shapes/>). Using the shape models and the density of  $1970 \text{ kg m}^{-3}$  for Alpha and  $2810 \text{ kg m}^{-3}$  for Beta, we can compute  $T_A^{lmn}$  and  $T_B^{lmn}$ . The purpose of this study is not to study the dynamical evolution details of the system, but to show the efficiency of our algorithm. So only one orbit is integrated. The initial orbital elements are

$$a = 2540.5 \text{ m}, \quad e = 0.01, \quad i = \Omega = \omega = 0^\circ, \quad M = 180^\circ$$

**Table 2** Computation time using our algorithm, for a 200 h integration of the orbit

| Order    | 2   | 3   | 4   | 5   | 6    | 7    | 8    | 9     |
|----------|-----|-----|-----|-----|------|------|------|-------|
| Time (s) | 0.2 | 0.7 | 2.1 | 6.6 | 17.3 | 42.6 | 99.3 | 206.0 |

**Fig. 3** The time history curves of the semi-major axis, the orbit eccentricity, and the orbit inclination with respect to the original orbital plane. The lower right is the orbit in space

The initial angular speeds in the body-fixed frame of Alpha (and Beta) are

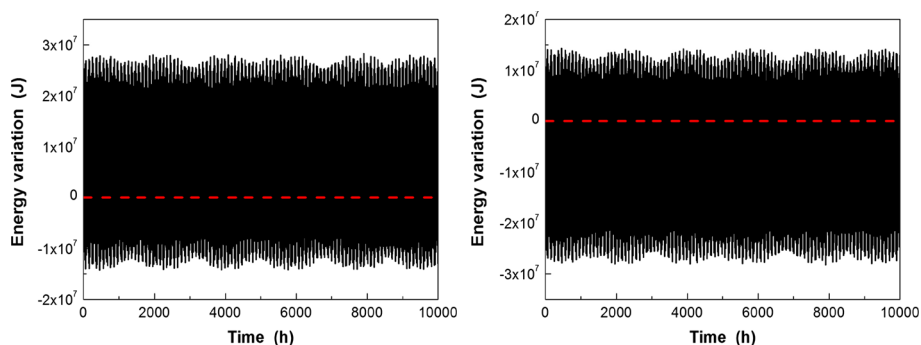
$$\Omega_A = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} 3125.4^\circ \cdot \text{day}^{-1}, \quad \Omega_B = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} 495^\circ \cdot \text{day}^{-1}$$

The (3–1–3) Euler angles of the body-fixed frame of Alpha (and Beta) with respect to the initial orbital plane are

$$\psi_A = 27.04^\circ, \quad \theta_A = 10^\circ, \quad \phi_A = -83.93^\circ \quad \psi_B = 0^\circ, \quad \theta_B = 0^\circ, \quad \phi_B = 180^\circ$$

For an integration time of 200h, Table 2 shows the computation time. The integrator is still the RKF7(8) integrator. The fixed time step is 200 s. The same single computer processor as the above one is used.

Compared with the several months using a single processor via the polyhedron approach with a 3rd order expansion (Scheeres et al. 2006), the superiority of our algorithm is obvious. Also, compared with the 36 days via the STF tensor approach with a 6th order expansion, the superiority of our algorithm is also obvious. Figure 3 shows the time history (10,000h, with a fixed time step of 200 s) of the semi-major axis, eccentricity, and orbit inclination



**Fig. 4** The variation curve of the total orbital energy (*left*) and the total rotational energy (*right*). The horizontal red line in these two frames indicates the total system energy variation

along with the orbit. Figure 4 shows the variations of the total orbit energy (sum of the orbital kinetic energy and the potential energy), the total rotational energy (sum of the rotational energies of the two bodies), and the total system energy with respect to their initial values. The potential is truncated at the sixth order. The total energy variation after the integration is  $-0.3851$  J. Compared with the total system energy around  $8.7670 \times 10^{10}$  J and considering the dissipative effects of the explicit RKF integrator, the total energy can be considered as conserved. Similar results hold for the total angular momentum of the system. The details are omitted here. From this figure, we know that there are large energy exchange between the rotational motion and the orbital motion due to the strong coupling between the two.

## 6 Planar two ellipsoids

The simplest F2RBP is the model composed of two rigid homogeneous spheres. The mutual potential is the same as for two point masses. A more complex model replaces one sphere with an ellipsoid (Scheeres 2007). The mutual potential between the two rigid bodies can still be given in a closed form in this case. However, when both rigid bodies are ellipsoids, no closed form of the mutual potential can be given. This model is very useful in studies of binary asteroid systems. Studies till now usually consider the mutual potential truncated at the 2nd order (Scheeres 2009; Jacobson and Scheeres 2011; Boldrin et al. 2016). For a qualitative description of the motion, this approximate model is proper because it captures the dominant non-spherical terms of both rigid bodies which declares its difference from the F2RBP composed of two spheres or one ellipsoid and a sphere. Nevertheless, it's important to describe quantitatively the accuracy of the 2nd order approximation. Meanwhile, it is also interesting to study whether any qualitative changes occur when mutual potentials of different orders are used. To the authors' knowledge, seldom work has been carried out on this comparison. Due to the computation efficiency of our algorithm, it is an ideal tool for such studies. As an example, we study one special case of the planar problem where the two ellipsoids are in contact and aligned with each other along their long axes (Scheeres 2009).

Assume the same homogenous density for the two ellipsoids with the semi axes as  $a_A \geq b_A \geq c_A$  and  $a_B \geq b_B \geq c_B$ . According to Tricarico (2008) (see also Macmillan 1958), the inertial integrals defined in Eqs. (16) and (20) are

$$T_*^{lmn} = 4\pi a_*^{l+1} b_*^{m+1} c_*^{n+1} \frac{\prod_{p=1}^{l/2} (2p-1) \prod_{q=1}^{m/2} (2q-1) \prod_{s=1}^{n/2} (2s-1)}{\prod_{u=1}^{(l+m+n)/2+2} (2u-1)} \quad (26)$$

where  $*$  indicates  $A$  or  $B$ . For ellipsoids,  $T_*^{lmn}$  is non-zero only when  $l, m, n$  are all even numbers. For the planar problem, computation time can be saved by rewriting Eq. (21) as

$$T_B^{lmn} = \sum_{i_1=0}^l \sum_{i_2=0}^m \frac{l!}{i_1!(l-i_1)!} \frac{m!}{i_2!(m-i_2)!} c_{11}^{i_1} c_{12}^{l-i_1} c_{21}^{i_2} c_{22}^{m-i_2} T_B^{(i_1+i_2)(l+m-i_1-i_2)n} \quad (27)$$

For the planar case, the EOMs expressed in Eq. (1) become (note that  $\dot{\mathbf{L}}'_B$  is replaced by  $\dot{\mathbf{L}}_B$  which is only possible for the planar case)

$$\begin{cases} \dot{\mathbf{R}} = \mathbf{R} \times \boldsymbol{\Omega}_A + \mathbf{V}, & \dot{\mathbf{V}} = \mathbf{V} \times \boldsymbol{\Omega}_A - \frac{1}{m} \frac{\partial U}{\partial \mathbf{R}} \\ \dot{\mathbf{L}}_A = \mathbf{M}_A, & \dot{\mathbf{L}}_B = \mathbf{M}'_B \end{cases} \quad (28)$$

where  $\mathbf{L}_A = I_Z^A \boldsymbol{\Omega}_A = I_Z^A \dot{\theta}_A \mathbf{e}_3$ ,  $\mathbf{L}_B = I_Z^B \boldsymbol{\Omega}_B = I_Z^B \dot{\theta}_B \mathbf{e}_3$ . The rotation matrixes are

$$C_A = \begin{bmatrix} \cos \theta_A & -\sin \theta_A & 0 \\ \sin \theta_A & \cos \theta_A & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad C_B = \begin{bmatrix} \cos \theta_B & -\sin \theta_B & 0 \\ \sin \theta_B & \cos \theta_B & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Define the relative angle as

$$\phi = \theta_B - \theta_A \quad (29)$$

to find

$$C = C_A^T C_B = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

For two ellipsoids aligned along their long axes and with a distance of  $R \geq a_A + a_B$  from each other's barycenter, the polar moment of inertia is (evaluated at the barycenter of the system)

$$I_p = \frac{m_A m_B}{m_A + m_B} R^2 + I_z^A + I_z^B = m R^2 + \frac{1}{5} m_A (a_A^2 + b_A^2) + \frac{1}{5} m_B (a_B^2 + b_B^2)$$

where  $m$  is already defined in Eq. (1). Since the two ellipsoids rotate synchronously with the orbital motion, denoting the angular velocity as  $\dot{\theta}_A$ , the total angular momentum of the system is

$$H = I_p \dot{\theta}_A \implies \dot{\theta}_A = \frac{H}{I_p}$$

and the energy of the system is

$$E = \frac{H^2}{2I_p} + U(\mathbf{R}, \phi)$$

where  $\mathbf{R}$  is the relative displacement between the barycenter of the two bodies, but expressed in the body fixed frame of  $A$ ,  $U$  is the mutual potential expressed by Eq. (17), and  $\phi$  is the relative geometry between the two bodies in the planar case defined by Eq. (29). In this case,  $\phi = 0$  because the two ellipsoids align along their long axis. According to Scheeres (2006

Proposition 7; 2012 Theorem 2), holding the angular momentum constant, the necessary condition for the configuration to be an equilibrium point is

$$0 = \frac{\partial E}{\partial R} = -\frac{H^2}{2I_p^2} \frac{\partial I_p}{\partial R} + \frac{\partial U}{\partial R}$$

Holding the relative direction between the two ellipsoids invariant as  $\phi = 0$ , the direction of the vector  $\mathbf{R}$  doesn't change. As a result, we have

$$\frac{\partial U}{\partial R} = G \sum_{n=0}^{\infty} \frac{n+1}{R^{n+2}} \tilde{U}_n$$

And the angular velocity of this system is

$$\dot{\theta}_A^2 = \frac{G}{mR} \sum_{n=0}^{\infty} \frac{n+1}{R^{n+2}} \tilde{U}_n \quad (30)$$

Truncated at the 2nd order of the potential, this result is already given in (Scheeres 2009). For the contact case, just substitute  $R = a_A + a_B$  into the above equation. Eq. (30) can also be obtained via the following formal way. Rewrite Eq. (28) as

$$\begin{cases} \ddot{X} = 2\dot{\theta}_A \dot{Y} + \ddot{\theta}_A Y + \dot{\theta}_A^2 X - \frac{1}{m} \frac{\partial U}{\partial X} \\ \ddot{Y} = -2\dot{\theta}_A \dot{X} - \ddot{\theta}_A X + \dot{\theta}_A^2 Y - \frac{1}{m} \frac{\partial U}{\partial Y} \\ \ddot{\theta}_A = \frac{M_z^A}{I_z^A}, \quad \ddot{\phi} = \frac{M_z^B}{I_z^B} - \frac{M_z^A}{I_z^A} \end{cases} \quad (31)$$

Further rewrite Eq. (31) using the polar coordinates as

$$\begin{cases} \ddot{R} = R(\dot{\theta} + \dot{\theta}_A)^2 - \frac{1}{m} \frac{\partial U}{\partial R} \\ \ddot{\theta} = -2\frac{\dot{R}}{R}(\dot{\theta} + \dot{\theta}_A) - \left(\frac{1}{I_z^A} + \frac{1}{mR^2}\right) \frac{\partial U}{\partial \theta} - \frac{1}{I_z^A} \frac{\partial U}{\partial \phi} \\ \ddot{\phi} = -\frac{1}{I_z^B} \frac{\partial U}{\partial \phi} - \frac{1}{I_z^A} \left(\frac{\partial U}{\partial \theta} + \frac{\partial U}{\partial \phi}\right) = -\frac{1}{I_z^A} \frac{\partial U}{\partial \theta} - \left(\frac{1}{I_z^A} + \frac{1}{I_z^B}\right) \frac{\partial U}{\partial \phi} \\ \ddot{\theta}_A = \frac{1}{I_z^A} \frac{\partial U}{\partial \theta} + \frac{1}{I_z^B} \frac{\partial U}{\partial \phi} \end{cases} \quad (32)$$

where  $R = \|\mathbf{R}\|$  and  $\theta$  is the quadrant angle of  $\mathbf{R}$ . For the contact equilibrium point, we have  $\ddot{\theta}_A = \ddot{\phi} = \dot{\phi} = 0$ ,  $\dot{\mathbf{R}} = \ddot{\mathbf{R}} = 0$ . As a result, Eq. (30) results from the first equation in Eq. (32). There is a relation between the system energy  $E$  and the total angular momentum  $H$

$$E - \frac{H^2}{2I_p} - U = \frac{1}{2I_p} \left[ I_z^A I_z^B \dot{\phi}^2 + mR^2 \left( I_z^A \dot{\theta} + I_z^B (\dot{\theta} - \dot{\phi}) \right)^2 \right] + \frac{1}{2} m \dot{R}^2 \quad (33)$$

This relation is called the zero-velocity surface in (Scheeres 2009), and appears in a similar form with different variables. Since the right hand side of Eq. (33) should be non-negative, this means

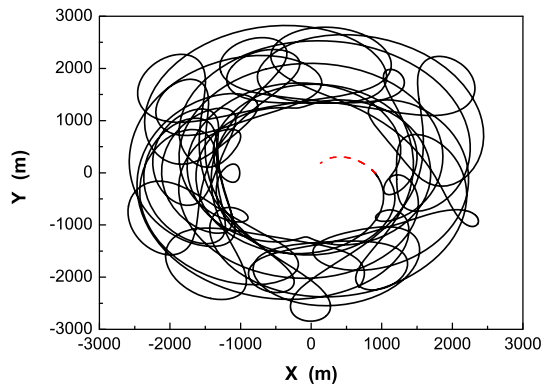
$$E - \frac{H^2}{2I_p(R)} - U(\mathbf{R}, \phi) \geq 0 \quad (34)$$

For every  $\mathbf{R}$ , there is a minimum value for the potential  $U(\mathbf{R}, \phi)$ . Denote it as  $U_{\min}(\mathbf{R})$ . From Eq. (34), we have

$$E - \frac{H^2}{2I_p(R)} - U_{\min}(R) \geq 0 \quad (35)$$

Equation (35) puts a limit on the possible region of the mutual orbit  $\mathbf{R}$ .

**Fig. 5** The outwards trajectory (solid line) and the inwards trajectory (dashed line). A 6th order potential is used



For the contact binary case in this study, the equilibrium point is unstable (see following Fig. 6 and related text). There are two kinds of trajectories after fission. One kind of trajectory follows the unstable invariant manifold outwards. If the system energy  $E < 0$  (i.e., if the system is Hill stable), the trajectory is confined by Eq. (35). It may come close to the equilibrium point again, but generally will never asymptotically approach that point unless the trajectory is exactly on a homoclinic orbit of the unstable equilibrium point. If the energy  $E > 0$ , the trajectory can be unconfined and may eventually escape (i.e., becomes Hill unstable). The other kind of trajectory follows the unstable invariant manifold inwards. It may also come close to the equilibrium point again, but still can never asymptotically approach that point unless it's also on a homoclinic orbit of the unstable equilibrium point. In practice, this kind of trajectory usually leads to re-impact of two binaries. Figure 5 shows the two kinds of trajectories in the body fixed frame of the ellipsoid A. The parameters for the two ellipsoids are

$$\begin{aligned} \alpha_A &= 1.0, & \beta_A &= 0.8, & \gamma_A &= 0.3, & R_A &= 500 \text{ m}, & \rho_A &= 2 \times 10^3 \text{ kg m}^{-3} \\ \alpha_B &= 1.0, & \beta_B &= 0.8, & \gamma_B &= 0.3, & R_B &= 400 \text{ m}, & \rho_B &= 2 \times 10^3 \text{ kg m}^{-3} \end{aligned} \quad (36)$$

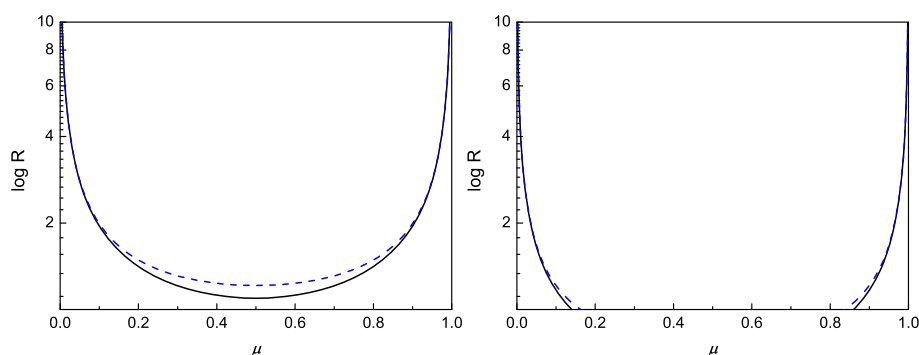
where  $\alpha_* = a_*/a_*$ ,  $\beta_* = b_*/a_*$ ,  $\gamma_* = c_*/a_*$ . In fact, the inwards trajectory shown in Fig. 5 is not appropriate due to two reasons: (1) the trajectory ignores the shapes of the ellipsoids. In practice, the two ellipsoids generally re-impact; (2) the mutual distance between the two ellipsoids is smaller than  $a_A + a_B$ , which causes the invalidity of the potential Eq. (18) based on the Legendre expansion Eq. (7). The potential is truncated at the 6th order for Fig. 5. The initial condition is given by Eq. (30). In order to perturb the initial condition from the equilibrium point,  $\dot{\theta}_A$  is multiplied by a factor 1.0001 for the outwards trajectory (also for Figs. 7, 8, 9, 10, 11, 12) and 0.9999 for the inwards trajectory.

Due to the absence of  $\theta_A$ , Eq. (32) is a 7 dimensional problem with variables as  $(R, \theta, \phi, \dot{R}, \dot{\theta}, \dot{\phi}, \dot{\theta}_A)$ . The variational equation can be expressed as

$$\delta \dot{X} = \tilde{A} \cdot \delta X \quad (37)$$

The eigenvalues of the matrix  $\tilde{A}$  are indicators of the linear stability of the equilibrium point. A linear restriction can be put on the variations  $\delta X$  by requiring the angular momentum of the system to be conserved, and as a result reducing the system to a 6 dimensional system—paralleling what was done in (Scheeres 2009). For our study, we consider the six variables

$$X' = (R, \theta, \phi, \dot{R}, \dot{\theta}, \dot{\phi})^T \quad (38)$$



**Fig. 6** *Left* the stability curve. Region below the curve is unstable and region above the curve is stable. *Right* the Hill stability curve. Region below the curve is Hill unstable and region above the curve is Hill stable. Unit of the ordinate in these two frames is  $R_A + R_B$ . The *solid lines* represent the results of the 2nd order potential, and the *dashed lines* represent the results of the 4th order potential

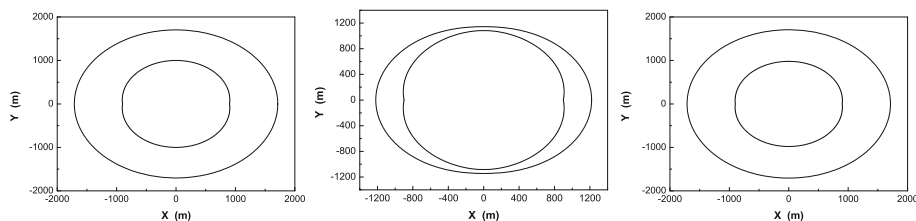
As a result, we have the variational equations as

$$\delta \dot{X}' = \tilde{A}' \cdot \delta X \quad (39)$$

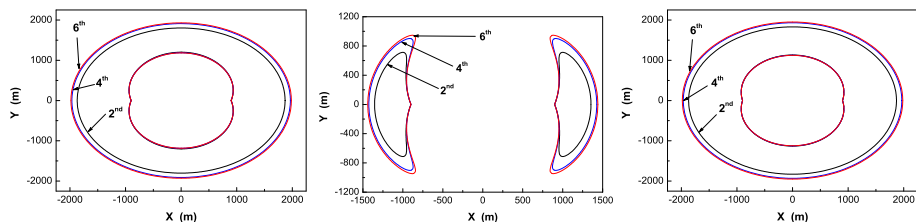
where the matrix  $\tilde{A}'$  is given in the Appendix 2 for a potential truncated at the 4th order. Using this equation, we are able to calculate the stability of the system by computing the eigenvalues of  $\tilde{A}'$ . In Scheeres (2009) it is shown that the contact equilibrium point is always unstable for the 2nd order potential. Evaluating the stability condition at higher orders we find that the instability of the contact problem doesn't change. If the distance  $R$  is increased from the contact value  $a_A + a_B$  while keeping angular momentum constant and changing the energy for the system to be in a relatively equilibrium, we eventually find a distance where the equilibrium point is stable (Scheeres 2009). The distance at which stability transitions from instability to stability depends on the mass fraction (which is defined as  $\mu = m_B / (m_A + m_B)$  in the following) and the shapes of the ellipsoids. To simplify our studies, we only consider the case where the two ellipsoids have the same shape, fixing  $\alpha_* = 1.0$ ,  $\beta_* = 0.3$ ,  $\gamma_* = 0.1$  (i.e., two very oblate and elongate ellipsoids) and varying  $R_A$ ,  $R_B$  (i.e., varying the mass ratio). The left frame of Fig. 6 shows the transition curve from instability to stability. The solid black line is for the 2nd order potential and the dashed blue line is for the 4th order potential. The right frame shows the corresponding curves for the Hill stability (i.e., when the total energy becomes positive). There is a difference in the curves when potential of different orders are used, but generally the difference is small.

In the following, we further compare the results when potentials of different orders are used. We focus on the trajectories after the two ellipsoids leave the unstable equilibrium configuration. We separate the comparison into two subcases: the  $E < 0$  case and the  $E > 0$  case. First, we study the  $E < 0$  case, where the mutual orbit is Hill stable. We start with two ellipsoids with a small  $J_{22}$  term. Fixing  $\alpha_A = \alpha_B = 1.0$ ,  $\beta_A = \beta_B = 0.99$ ,  $\gamma_A = \gamma_B = 0.98$ , from left to right, Fig. 7 shows the ZVCs of  $R_A = 500$ ,  $R_B = 400$  m ( $\mu \approx 0.34$ ),  $R_A = 450$ ,  $R_B = 450$  m ( $\mu = 0.50$ ), and  $R_A = 400$ ,  $R_B = 500$  m ( $\mu \approx 0.66$ ). It seems that the feasible region after fission doesn't change with the variation of the mass ratio. Figure 7 is for the 2nd order potential. Since non-spherical terms are very small, considering higher order terms doesn't make any obvious change to the zero velocity curves (ZVCs). One remark is that the left frame and the right frame in Fig. 7 shows the same case, but the ZVC is shown

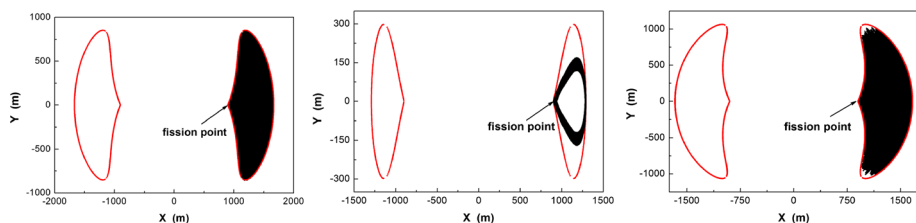




**Fig. 7** ZVCs for small  $\alpha_* - \beta_*$  after fission, for the 2nd order potential. Left  $\mu \approx 0.34$ ; middle  $\mu = 0.50$ ; right  $\mu \approx 0.66$



**Fig. 8** ZVCs for moderate  $\alpha_* - \beta_*$  after fission. Black line is for the 2nd order potential, blue line is for the 4th order potential and the red line is for the 6th order potential. From left to right, the mass ratios are same as Fig. 7

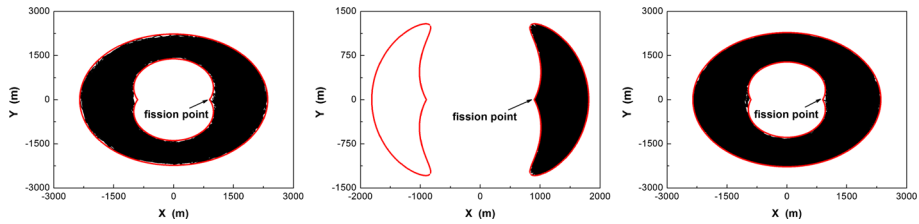


**Fig. 9** ZVCs for large  $\alpha_* - \beta_*$  after fission, for the 2nd order potential. The red curve is the ZVC, and the black curve is the trajectory after fission, integrated for 10,000 Keplerian orbital periods. From left to right, the mass ratios are same as Fig. 7

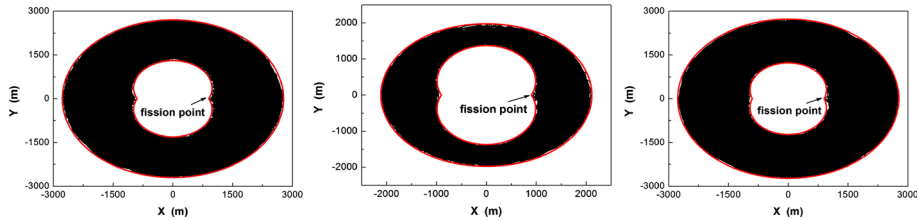
in different coordinates. The left one shows the ZVC in the body fixed frame of the larger primary while the right one shows the ZVC in the body fixed frame of the smaller primary.

Next consider two ellipsoids with a moderate  $J_{22}$  term. Fixing  $\alpha_A = \alpha_B = 1.0$ ,  $\beta_A = \beta_B = 0.9$ ,  $\gamma_A = \gamma_B = 0.8$ , from left to right, Fig. 8 shows the ZVCs for mass ratios same as those in Fig. 7. Black lines are for the 2nd order potential, blue lines are for the 4th order potential and red lines are for the 6th order potential, and we note that considering higher order terms will not qualitatively change the ZVCs. Compared with Fig. 7 we see that for two ellipsoids with comparable masses to each other (i.e.,  $\mu \sim 0.5$ ), there is a qualitative change in the feasible region after fission for a changing  $J_{22}$ .

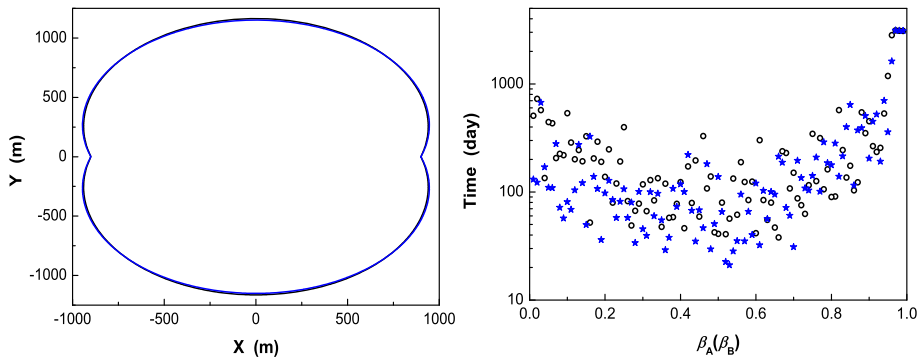
Last, we consider ellipsoids with a large  $J_{22}$  term. Fixing  $\alpha_A = \alpha_B = 1.0$ ,  $\beta_A = \beta_B = 0.5$ ,  $\gamma_A = \gamma_B = 0.3$ , from left to right, Figs. 9, 10 and 11 show the ZVCs for the same mass ratios as the above two figures. Figure 9 is for the 2nd order potential, Fig. 10 is for the 4th order potential, and Fig. 11 is for the 6th order potential. In these figures, red lines are the ZVCs and black lines are the trajectories after fission, integrated for about 10,000 Keplerian orbital periods. Obviously, there is a qualitative difference in the ZVCs when potentials of different orders are used. Potentials of lower orders may restrict trajectories of the secondary



**Fig. 10** Same as Fig. 9, but for the 4th order potential



**Fig. 11** Same as Fig. 9, but for the 6th order potential

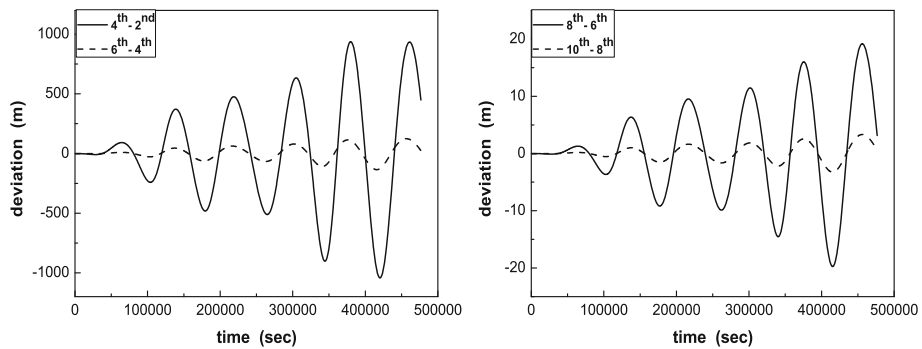


**Fig. 12** Left ZVCs for the Hill unstable case. Right escape time with respect to the parameter  $\beta_*$

after fission to be in libration with the primary (McMahon and Scheeres 2013), which is clearly not the truth when potentials of higher order are considered. As a result, to avoid being misled to a wrong description of the trajectories after fission, potentials of high orders are necessary. Usually, a potential truncated at the forth order can avoid this misleading.<sup>1</sup> However, judging from the middle frames in Figs. 9, 10 and 11, when the mass ratio is close to 0.5, potentials of orders higher than 4 may be required.

Next, we focus on the  $E > 0$  case. Although  $E > 0$  is only a necessary condition for the Hill instability, our numerical simulations show that nearly all the trajectories after fission escape. There is no outer boundary for the ZVCs but the inner boundary still exists. Fixing  $\alpha_A = \alpha_B = 1.0$ ,  $\beta_A = \beta_B = 0.5$ ,  $\gamma_A = \gamma_B = 0.3$ , the left frame of Fig. 12 shows the inner boundaries for the 2nd (black) and the 4th (blue) order potential. The ZVCs nearly coincide with each other when potentials of different orders are used. As an example, we carried out the computations shown in the right frame of Fig. 12. Fixing the mass ratio

<sup>1</sup> Potential truncated at the 4th order is given in Appendix 1, along with the force and the torque. Note that these formulae are only valid for the planar case.



**Fig. 13** Difference between the integrated orbits ( $x$  component) with the same initial condition when potentials of different orders are used

( $R_A = 600$ ,  $R_B = 300$  m, corresponding to  $\mu \approx 0.11$ ) and changing the shape of the ellipsoids, we record the time when the secondary ellipsoid reaches a distance of 100 radii of the primary ellipsoid. To simplify our study, we fix  $\alpha_* = 1$ ,  $\gamma_* = 0.3$  and vary  $\beta_*$ . The unfilled black circle corresponds to the 2nd order potential, and the filled blue star corresponds to the 4th order potential. For a specific value of  $\beta_*$ , the escape time is obviously different when potentials of different orders are used, but there is no obvious difference in the distribution of the escaping time with respect to  $\beta_*$  between the 2nd and the 4th order result. This suggests us that we can carry out some statistical studies even with the lowest 2nd order potential (Jacobson and Scheeres 2011; Boldrin et al. 2016). Judging from the shape of the ZVCs and the study in the right frame of Fig. 12, there should be no qualitative difference for the  $E > 0$  case when potentials of different orders are used. However, we should be careful about this conclusion because only very limited examples are computed here and further studies are required.

As a concluding study for this section, we provide a quantitative comparison between the results of different orders. The comparison is not done for the contact case, as those motions are chaotic which could amplify the differences between force models. As a result, we choose a regular orbit. The two ellipsoids are still the ones given by Eq. (36). They are initially separated from each other at a distance of  $2(a_A + a_B)$  from their barycenter, with an initial alignment along their long axes. The initial rotation rate  $\dot{\theta}_A$  is same as the Keplerian orbital angular velocity of the two ellipsoids which is computed from the simple two-body relation. The initial rotation angle of the body  $A$  is  $\theta_A = 0$ . The initial relative rotation angle is  $\phi = 0$  and the initial relative rotation speed is  $\dot{\phi} = 0$ . The parameters of the two ellipsoids are same as Fig. 5. Taking the  $x$  component as an example, Fig. 13 shows the differences between the  $n$ th order and the  $(n + 2)$ th order of the potential with the time going. Different from the above figures, the orbit in Fig. 13 is shown in the inertial frame centered at the barycenter of the body  $A$ . Initial conditions of each orbit in Fig. 13 are same as each other, but only integrated in potentials truncated at different orders. Obviously, with increasing truncation order  $n$ , the difference becomes smaller.

## 7 Conclusion and discussion

In the first part of the paper, the mutual potential along with the force and the torque for the F2RBP are given with explicit formulae. The steps to these formulae are plain and

easy. The results are valid for two rigid bodies of arbitrary shapes and mass distributions. Numerical simulations and comparisons with previous studies demonstrate the superiority of the algorithm. As a result, we conclude that it is advantageous to use the new algorithm in practical studies.

Qualitatively, our algorithm is equivalent to that of Paul (1988) and Tricarico (2008). However, instead of 6 open summations in their work, only one open summation (the truncation order of the potential) appears in our formulae. At the cost of three more summation signs (but no computation cost), the orbital component (expressed by  $\mathbf{e} = \mathbf{R}/R$ ), the shape dependent parts of  $A$  and  $B$  (expressed by  $T_A^{lmn}$  and  $T_B^{lmn}$ ), and the relative attitude coupling at different orders (expressed by the matrix  $C$  and Eq. (21)) are neatly separated from each other. The recursive formulae expressed by Eqs. (9) and (13) make the programming work much easier. In fact, substituting Eqs. (21) into (18), we can get rid of the intermediate variables  $T_B^{lmn}$  and arrive at the form given by Paul and Tricarico. Taking the expression given by Tricarico as an example, Eq. (40) (same as Eq. (7) in Tricarico (2008)) gives the formulae for the potential

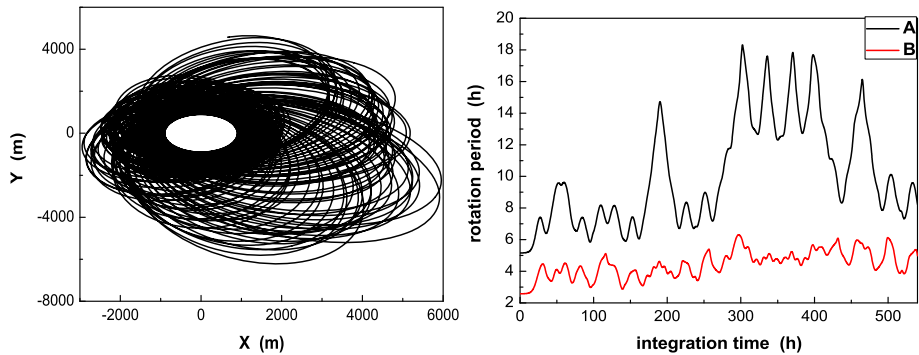
$$\begin{aligned}
 U_{12} = & G \frac{M_1 M_2}{R} \sum_{i_1=0}^{\infty} \sum_{j_1=0}^{\infty} \sum_{k_1=0}^{\infty} \sum_{i_2=0}^{\infty} \sum_{j_2=0}^{\infty} \sum_{k_2=0}^{\infty} (-1)^{i_1+j_1+k_1} \\
 & \times \frac{N_{1,i_1 j_1 k_1}}{i_1! j_1! k_1!} \frac{N_{2,i_2 j_2 k_2}}{i_2! j_2! k_2!} \times \left(\frac{r_0}{R}\right)^{i_1+j_1+k_1+i_2+j_2+k_2} \\
 & \times \sum_{i_3=0}^{i_2} \sum_{j_3=0}^{j_2} \sum_{k_3=0}^{k_2} \sum_{i_4=0}^{i_3} \sum_{j_4=0}^{j_3} \sum_{k_4=0}^{k_3} \binom{i_2}{i_3} \binom{j_2}{j_3} \binom{k_2}{k_3} \binom{i_3}{i_4} \binom{j_3}{j_4} \binom{k_3}{k_4} \\
 & \times l_x^{i_4} m_x^{i_3-i_4} n_x^{i_2-i_3} l_y^{j_4} m_y^{j_3-j_4} n_y^{j_2-j_3} l_z^{k_4} m_z^{k_3-k_4} n_z^{k_2-k_3} \\
 & \times \sum_{L=0}^{i_5} \sum_{M=0}^{j_5} \sum_{N=0}^{k_5} t_{i_5 j_5 k_5 L M N} \frac{\xi_1^L \eta_1^M \zeta_1^N}{R^{L+M+N}} \quad (40)
 \end{aligned}$$

where  $\xi_1, \eta_1, \zeta_1$  are  $e_x, e_y, e_z$  defined as  $\mathbf{R}/R$ , and

$$\begin{aligned}
 i_5 &= i_1 + i_4 + j_4 + k_4 \\
 j_5 &= j_1 + i_3 - i_4 + j_3 - j_4 + k_3 - k_4 \\
 k_5 &= k_1 + i_2 - i_3 + j_2 - j_3 + k_2 - k_3
 \end{aligned}$$

It's a plain fact that the coefficients  $t_{i_5 j_5 k_5 L M N}$  in Eq. (40) are combinations of the coefficients  $a_{(i_1 i_2 i_3)(i_4 i_5 i_6)}^k, b_{(j_1 j_2 j_3)(j_4 j_5 j_6)}^{n-k}$  and  $t_k^n$  in our expressions. The recursive relation for them is given by Eq. (2) in (Tricarico 2008). In computing the potential, our expression and the expression given by Paul and Tricarico should be identical to each other. The difference comes when taking partial derivatives with respect to the position vector  $\mathbf{R}$ . In our expression, the intermediate variables  $T_B^{lmn}$  can be stored and reused in Eq. (22). But in Paul's or Tricarico's expression, they have to be recomputed again (see Eq. (8) in Tricarico 2008).

Using the algorithm as a tool, the second part of the paper is devoted to the planar problem of two ellipsoids. This model can be used to simulate the contact fission of asteroids. Previous studies were limited to the 2nd order potential (Scheeres 2009; Jacobson and Scheeres 2011; Boldrin et al. 2016). Our study aims to answer the question whether there are any qualitative differences when higher order terms of the potential are considered. The problem is separated into the case where the system is Hill stable ( $E < 0$ ) and when it is not necessarily Hill stable ( $E > 0$ ). The 2nd order potential is good enough for a qualitative description of the



**Fig. 14** *Left* the trajectory after fission for about 10,00 Keplerian orbital periods. The mass ratio is  $\mu \approx 0.11$ . *Right* the rotation curves of the two ellipsoids. The potential for these two frames is truncated at the 6th order

trajectories after fission. For the  $E < 0$  case where the trajectories are bounded by the ZVCs, it is a bit more complicated. For ellipsoids with a small  $J_{22}$  term, the 2nd order potential is good enough. For ellipsoids with a very large  $J_{22}$  term, potentials truncated at high orders ( $\geq 4$ ) are necessary in qualitatively describing the motion after fission.

Due to its fast computational speed, this algorithm may have several applications for future studies. Suppose we have accurate observation data of a binary system, we can determine the inertia integrals  $T_A^{lmn}$  and  $T_B^{lmn}$  from these observations, and from the inertia integrals, we can recover the spherical harmonics. If we also have the shape information of the binaries, we can then estimate the mass distribution of each component in this binary system (Takahashi and Scheeres 2014).

Another potential application is the tidal dissipation theory for binary asteroids. The classical tidal theory usually only considers the second order tidal potential (Murray and Dermott 1999; Ferraz-Mello et al. 2008). This is acceptable because the magnitude of higher order terms rapidly reduces when the two binaries are far away from each other. But for very close-in binary asteroids, there are at least two new things to be considered:

- (1) The effects of higher order terms. For binary asteroids with comparable masses to each other (the  $E < 0$  case), their relative motion after fission is generally confined in the proximity of each other by the ZVCs (see Figs. 7, 8, 9, 10 and 11). In this case, the magnitude of high order terms of the tidal potential reduces much slower and should be considered. Even for the  $E > 0$  case, if the non-spherical terms are small, the secondary may spend quite a long time in the proximity of the primary before it escapes, as shown in the left frame of Fig. 14 ( $\alpha_* = 1.0$ ,  $\beta_* = 0.96$ ,  $\gamma_* = 0.9$ ,  $R_A = 600$ ,  $R_B = 300$  m). In fact, the role of high order terms in the tidal dissipation theory was already studied by Taylor and Margot for the case of two spheres (2010) and the case of one sphere and one ellipsoid (2014). It would be more difficult if the shape irregularities of the asteroids are considered.
- (2) For real asteroids, changes in their rotation rates can cause relative deformations. For classical problems in Celestial Mechanics, this change is negligible because the change is usually extremely slow. The energy is mainly dissipated by the tides. However, for binary asteroids, the rotation speed of the two ellipsoids changes quickly when they are close to each other due to the spin-orbit coupling, as shown in the right frame of Fig. 14. Changes in the deformations caused by the variation in the rotation speed can also dissipate energy. Actually, changes in the rotation speed may cause the redistribution

of masses on the asteroids (Sánchez and Scheeres 2012; Scheeres 2015), thus violating the assumption of elastic deformation for the classical tidal theory. However, it's still of interest to answer the question that whether this energy dissipation mechanism is important and how it works.

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## Appendix 1: explicit expressions of the 4th order potential for the planar two ellipsoids problem

We provide formulae for the 4th order potential here. In these formulae, we define

$$\delta = \theta - \phi$$

where  $\theta$  is the quadrant angle of the position vector  $\mathbf{R}$  in the body-fixed frame of A, and  $\phi$  is defined in Sect. 5. The relative geometry of the system is defined in Fig. 15.

In the following, the mutual potential is defined by Eq. (19) where

$$\tilde{U}_0 = T_A^{000} T_B^{000}$$

$$\tilde{U}_2 = A_1 + A_2 \cos 2\theta + A_3 \cos 2\delta$$

$$\tilde{U}_4 = B_1 + B_2 \cos 2\theta + B_3 \cos 4\theta + B_4 \cos 2\delta + B_5 \cos 4\delta + B_6 \cos 2\phi + B_7 \cos(2\theta + 2\delta)$$

where

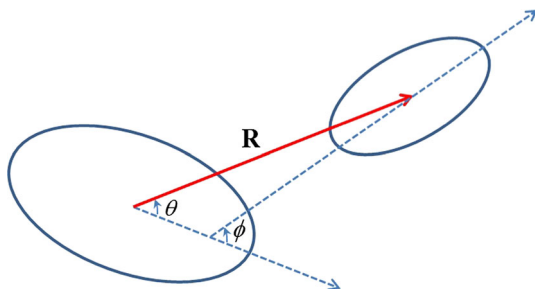
$$A_1 = \frac{1}{4} (T_B^{200} + T_B^{020} - 2T_B^{002}) T_A^{000} + \frac{1}{4} (T_A^{200} + T_A^{020} - 2T_A^{002}) T_B^{000}$$

$$A_2 = \frac{3}{4} (T_A^{200} - T_A^{020}) T_B^{000},$$

$$A_3 = \frac{3}{4} (T_B^{200} - T_B^{020}) T_A^{000}$$

$$B_1 = \frac{9}{64} T_B^{000} \left( T_A^{400} + T_A^{040} + \frac{8}{3} T_A^{004} + 2T_A^{220} - 8T_A^{022} - 8T_A^{202} \right) \\ + \frac{9}{64} T_A^{000} \left( T_B^{400} + T_B^{040} + \frac{8}{3} T_B^{004} + 2T_B^{220} - 8T_B^{022} - 8T_B^{202} \right)$$

**Fig. 15** A picture illustrating the relative geometry of the system and the variables used



$$\begin{aligned}
& -\frac{9}{8}T_B^{002}(T_A^{200} + T_A^{020} - T_A^{002}) - \frac{9}{8}T_A^{002}(T_B^{200} + T_B^{020} - T_B^{002}) \\
& + \frac{9}{16}(T_A^{200} + T_A^{020})(T_B^{200} + T_B^{020}) \\
B_2 &= \frac{5}{16}T_B^{000}(T_A^{400} - T_A^{040} + 6T_A^{022} - 6T_A^{202}) \\
& + \frac{15}{16}(T_A^{200} - T_A^{020})(T_B^{200} + T_B^{020} - 2T_B^{002}) \\
B_3 &= \frac{35}{16}T^{00}(T_A^{400} + T_A^{040} - 6T_A^{220}) \\
B_4 &= \frac{5}{16}T_A^{000}(T_B^{400} - T_B^{040} + 6T_B^{022} - 6T_B^{202}) \\
& + \frac{15}{16}(T_B^{200} - T_B^{020})(T_A^{200} + T_A^{020} - 2T_A^{002}) \\
B_5 &= \frac{35}{64}T_A^{000}(T_B^{400} + T_B^{040} - 6T_B^{220}) \\
B_6 &= \frac{9}{32}(T_A^{200} - T_A^{020})(T_B^{200} - T_B^{020}) \\
B_7 &= \frac{105}{32}(T_A^{200} - T_A^{020})(T_B^{200} - T_B^{020})
\end{aligned}$$

For the 2nd order potential, it's obvious that  $U_{\min}(\mathbf{R})$  corresponds to the maximum value of  $\tilde{U}_2$ . This means  $\delta = 0$  (i.e.,  $\theta = \phi$ ). As a result, we have

$$U_{\min}(\mathbf{R}) = -G \left( \frac{\tilde{U}_0}{R} + \frac{\tilde{U}_2^{\max}}{R^3} \right)$$

where

$$\tilde{U}_2^{\max} = A_1 + A_2 \cos 2\theta + A_3$$

For the potential truncated at the 4th order,  $U_{\min}(\mathbf{R})$  usually doesn't exactly happen at  $\delta = 0$ , especially for small values of  $R$ . The exact value of  $\phi$  for  $U_{\min}(\mathbf{R})$  can be obtained by the following condition

$$\frac{\partial U(\mathbf{R}, \phi)}{\partial \phi} = 0$$

Usually there are two solutions to this equation. The minimum value happens for  $\delta$  close to  $\theta$ .

Equations of motion are already given by Eq. (30). The forces are given by

$$\begin{aligned}
\frac{\partial U}{\partial X} &= G \left[ X \left( \frac{\tilde{U}_0}{R^3} + \frac{3\tilde{U}_2}{R^5} + \frac{5\tilde{U}_4}{R^7} \right) - \left( \frac{1}{R^3} \frac{\partial \tilde{U}_2}{\partial \theta} + \frac{1}{R^5} \frac{\partial \tilde{U}_4}{\partial \theta} \right) \frac{\partial \theta}{\partial X} \right] \\
\frac{\partial U}{\partial Y} &= G \left[ Y \left( \frac{\tilde{U}_0}{R^3} + \frac{3\tilde{U}_2}{R^5} + \frac{5\tilde{U}_4}{R^7} \right) - \left( \frac{1}{R^3} \frac{\partial \tilde{U}_2}{\partial \theta} + \frac{1}{R^5} \frac{\partial \tilde{U}_4}{\partial \theta} \right) \frac{\partial \theta}{\partial Y} \right]
\end{aligned}$$

where

$$\frac{\partial \theta}{\partial X} = -\frac{Y}{R^2}, \quad \frac{\partial \theta}{\partial Y} = \frac{X}{R^2}$$

The torque  $M_z^B$  is given by

$$M_z^B = -\frac{\partial U}{\partial \theta_B} = -\frac{\partial U}{\partial \phi} \frac{\partial \phi}{\partial \theta_B} = -\frac{\partial U}{\partial \phi} = G \left[ \frac{1}{R^3} \frac{\partial \tilde{U}_2}{\partial \phi} + \frac{1}{R^5} \frac{\partial \tilde{U}_4}{\partial \phi} \right]$$

and the torque  $M_z^A$  is given by

$$\begin{aligned} M_z^A &= -\frac{\partial U}{\partial \theta_A} = -\frac{\partial U}{\partial \theta} \frac{\partial \theta}{\partial \theta_A} - \frac{\partial U}{\partial \phi} \frac{\partial \phi}{\partial \theta_A} = \frac{\partial U}{\partial \theta} + \frac{\partial U}{\partial \phi} = \mathbf{R} \times \frac{\partial U}{\partial \mathbf{R}} - M_z^B \\ &= -G \left[ \frac{1}{R^3} \frac{\partial \tilde{U}_2}{\partial \theta} + \frac{1}{R^5} \frac{\partial \tilde{U}_4}{\partial \theta} \right] - G \left[ \frac{1}{R^3} \frac{\partial \tilde{U}_2}{\partial \phi} + \frac{1}{R^5} \frac{\partial \tilde{U}_4}{\partial \phi} \right] \end{aligned}$$

We have

$$\begin{aligned} \frac{\partial \tilde{U}_2}{\partial \theta} &= -2A_2 \sin 2\theta - 2A_3 \sin 2\delta \\ \frac{\partial \tilde{U}_2}{\partial \phi} &= 2A_3 \sin 2\delta \frac{\partial \tilde{U}_4}{\partial \theta} = -2B_2 \sin 2\theta \\ &\quad - 4B_3 \sin 4\theta - 2B_4 \sin 2\delta - 4B_5 \sin 4\delta - 4B_7 \sin (2\theta + 2\delta) \\ \frac{\partial \tilde{U}_4}{\partial \phi} &= 2B_4 \sin 2\delta + 4B_5 \sin 4\delta - 2B_6 \sin 2\phi + 2B_7 \sin (2\theta + 2\delta) \end{aligned}$$

The accuracy of these formulae is checked by a handmade program. Integrated orbits using these formulae are compared with the ones integrated by using Eq. (21) (or. Eq. (27)) which needs the intermediate variables  $T_B^{lmn}$ . Details of these comparisons are omitted. These formulae should be useful for future studies. Compared with the 2nd order potential, increase of the computation cost is acceptable.

One remark is made here. Utilizing the fact that

$$\theta = -\phi_1, \quad \delta = -\phi_2$$

The second order potential is identical to that given in (Scheeres 2009) which used a different set of variables. For researchers who are interested in this set of variables, the 4th order potential can be easily obtained from the formulae given here.

## Appendix 2: the matrix $\tilde{A}'$

From Eq. (32), neglecting the obvious zero terms for the contact equilibrium, we have the following variational equations

$$\begin{cases} \delta \ddot{R} = (\dot{\theta} + \dot{\theta}_A)^2 \delta R + 2R (\dot{\theta} + \dot{\theta}_A) \delta \dot{\theta} + 2R (\dot{\theta} + \dot{\theta}_A) \delta \dot{\theta}_A - \frac{1}{m} U_{RR} \delta R \\ \delta \ddot{\theta} = - \left[ \left( \frac{1}{I_z^A} + \frac{1}{mR^2} \right) U_{\theta\theta} + \frac{1}{I_z^A} U_{\theta\phi} \right] \delta \theta - \left[ \left( \frac{1}{I_z^A} + \frac{1}{mR^2} \right) U_{\theta\phi} + \frac{1}{I_z^A} U_{\phi\phi} \right] \delta \phi - \frac{2(\dot{\theta} + \dot{\theta}_A)}{R} \delta \dot{R} \\ \delta \ddot{\phi} = - \left[ \left( \frac{1}{I_z^A} + \frac{1}{I_z^B} \right) U_{\theta\phi} + \frac{1}{I_z^A} U_{\theta\theta} \right] \delta \theta - \left[ \left( \frac{1}{I_z^A} + \frac{1}{I_z^B} \right) U_{\phi\phi} + \frac{1}{I_z^A} U_{\theta\phi} \right] \delta \phi \end{cases}$$

Utilizing the fact that the total angular momentum of the system conserves, we have

$$\delta \dot{\theta}_A = -\frac{2mR (\dot{\theta}_A + \dot{\theta})}{I_z} \delta R - \frac{mR^2}{I_z} \delta \dot{\theta} - \frac{I_z^B}{I_z} \delta \dot{\phi}$$



Substituting the relation to the above equation, we have

$$\begin{cases} \delta \ddot{R} = a_{41} \delta R + a_{45} \delta \dot{\theta} + a_{46} \delta \dot{\phi} \\ \delta \ddot{\theta} = a_{52} \delta \theta + a_{53} \delta \phi + a_{54} \delta \dot{R} \\ \delta \ddot{\phi} = a_{62} \delta \theta + a_{63} \delta \phi \end{cases}$$

where

$$\begin{aligned} a_{41} &= \dot{\theta}_A^2 \left( 1 - \frac{4mR^2}{I_z} \right) - \frac{1}{m} U_{RR}, \\ a_{45} &= 2R\dot{\theta}_A \left( 1 - \frac{mR^2}{I_z} \right), \\ a_{46} &= -\frac{2R\dot{\theta}_A I_z^B}{I_z}, \\ a_{52} &= -\left[ \left( \frac{1}{I_z^A} + \frac{1}{mR^2} \right) U_{\theta\theta} + \frac{1}{I_z^A} U_{\theta\phi} \right], \\ a_{53} &= -\left[ \left( \frac{1}{I_z^A} + \frac{1}{mR^2} \right) U_{\theta\phi} + \frac{1}{I_z^A} U_{\phi\phi} \right], \\ a_{54} &= -\frac{2\dot{\theta}_A}{R}, \\ a_{62} &= -\left[ \left( \frac{1}{I_z^A} + \frac{1}{I_z^B} \right) U_{\theta\phi} + \frac{1}{I_z^A} U_{\theta\theta} \right], \\ a_{63} &= -\left[ \left( \frac{1}{I_z^A} + \frac{1}{I_z^B} \right) U_{\phi\phi} + \frac{1}{I_z^A} U_{\theta\phi} \right] \end{aligned}$$

The matrix  $\tilde{A}'$  is a plain fact from the above equation. For the equilibrium case studied in this paper (i.e., two ellipsoids aligning with each other along their long axes), we have

$$\begin{aligned} \dot{\theta}_A^2 &= \frac{G}{m} \left[ \frac{\tilde{U}_0}{R^3} + \frac{3(A_1 + A_2 + A_3)}{R^5} + \frac{5(B_1 + B_2 + B_3 + B_4 + B_5 + B_6 + B_7)}{R^7} \right] \\ U_{RR} &= -G \left[ \frac{2\tilde{U}_0}{R^3} + \frac{12(A_1 + A_2 + A_3)}{R^5} + \frac{30(B_1 + B_2 + B_3 + B_4 + B_5 + B_6 + B_7)}{R^7} \right] \\ U_{\theta\theta} &= 4G \left[ \frac{(A_2 + A_3)}{R^3} + \frac{(B_2 + 4B_3 + B_4 + 4B_5 + B_6 + 4B_7)}{R^5} \right] \\ U_{\theta\phi} &= -4G \left[ \frac{A_3}{R^3} + \frac{(B_4 + 4B_5 + 2B_7)}{R^5} \right] \\ U_{\phi\phi} &= 4G \left[ \frac{A_3}{R^3} + \frac{(B_4 + 4B_5 + B_6 + B_7)}{R^5} \right] \end{aligned}$$

The eigenvalues of the system satisfies the following equation

$$\lambda^6 - (a_{41} + a_{52} + a_{63} + a_{45}a_{54})\lambda^4 + (a_{41}a_{63} + a_{52}a_{63} + a_{41}a_{52} - a_{53}a_{62} + a_{45}a_{54}a_{63} - a_{46}a_{62}a_{54})\lambda^2 + (a_{53}a_{62}a_{41} - a_{41}a_{52}a_{63}) = 0$$

From this equation, we compute the eigenvalues and generate the left frame of Fig. 6.

For the equilibrium case studied in this paper, the system energy is

$$E = \frac{H^2}{2I_p} + U = \frac{1}{2}(mR^2 + I_z^A + I_z^B) \frac{G}{m} \left[ \frac{\tilde{U}_0}{R^3} + \frac{3D_1}{R^5} + \frac{5D_2}{R^7} \right] - G \left[ \frac{\tilde{U}_0}{R} + \frac{D_1}{R^3} + \frac{D_2}{R^5} \right]$$

where

$$\begin{aligned} D_1 &= A_1 + A_2 + A_3 \\ D_2 &= B_1 + B_2 + B_3 + B_4 + B_5 + B_6 + B_7 \end{aligned}$$

The condition  $E = 0$  gives out the following equation

$$\begin{aligned} 5D_2 \left( I_z^A + I_z^B \right) \left( \frac{1}{R^2} \right)^3 &+ \left[ 3D_1 \left( I_z^A + I_z^B \right) + 3mD_2 \right] \left( \frac{1}{R^2} \right)^2 \\ &+ \left[ \left( I_z^A + I_z^B \right) \tilde{U}_0 + mD_1 \right] \left( \frac{1}{R^2} \right) - m\tilde{U}_0 = 0 \end{aligned}$$

From this equation, we compute the  $R_H \geq 1$  in the right frame of Fig. 6.

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